

Work Breakdown Structure (WBS)

Community Tool & Equipment Library — Problem 52

1. Requirements Engineering

- 1.1 Review problem statement
- 1.2 Identify actors
- 1.3 Define functional requirements
- 1.4 Define non-functional requirements
- 1.5 Define acceptance criteria
- 1.6 Maintain requirements traceability

2. System Analysis & Design

- 2.1 Use-case analysis
- 2.2 Main/alternate/exception flows
- 2.3 Architectural design
- 2.4 Data and service decomposition
- 2.5 Interface considerations

3. Core System Development

- 3.1 User management
- 3.2 Tool inventory and search/filter
- 3.3 Reservation and checkout
- 3.4 Return and damage assessment
- 3.5 Deposit and late-fee processing
- 3.6 Due-date and overdue notifications

4. Project Management

- 4.1 GitHub repository management
- 4.2 Jira project/epic/story/task management
- 4.3 Sprint/backlog tracking
- 4.4 Commit and documentation management

5. AI-Assisted Development

- 5.1 Select a suitable coding task
- 5.2 Generate/assist code with GitHub Copilot
- 5.3 Review generated code
- 5.4 Test the generated/modified code

- 5.5 Commit the accepted implementation
- 5.6 Capture Copilot evidence

6. Software Testing & Bug Fixing

- 6.1 Obtain the faculty-provided game repository
- 6.2 Read its README and four test cases
- 6.3 Execute the supplied tests
- 6.4 Identify the failing behavior/bug
- 6.5 Use vibe coding to prepare a fix/patch
- 6.6 Apply and review the patch
- 6.7 Retest all relevant test cases
- 6.8 Push the fixed repository
- 6.9 Document before/after evidence

7. Submission & Documentation

- 7.1 Prepare SRS
- 7.2 Prepare WBS
- 7.3 Organize architecture evidence
- 7.4 Organize GitHub/Jira evidence
- 7.5 Organize Copilot evidence
- 7.6 Organize testing evidence
- 7.7 Update final README with links to all six folders